WORKED SOLUTION

STEP III 2008, Question 10

Mechanics • Hooke's Law

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/08-s3-q10

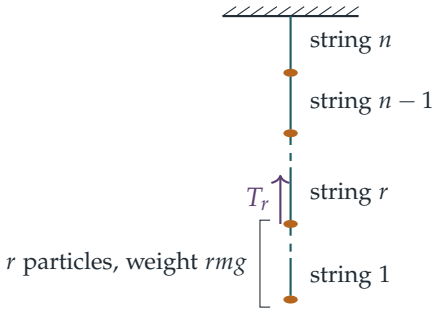
The key idea

Each short string holds up everything that hangs below it, so its tension is easy to write down; Hooke's law then gives its extension and its stored energy, and the totals are standard sums of r and r^2. The rope is the same picture with the mass spread out: chop it into n equal pieces, treat each piece's mass as a particle at its lower end, and let $n \to \infty$. The sums become integrals, and the factor $\frac{1}{2}$ in the length appears because, on average, a piece of rope supports half the rope's weight.

Solution

The string of particles

Model the short strings as light elastic strings obeying Hooke's law, and the masses as particles. Number the strings from the bottom, as in the question, so string r has r particles below it: its own particle and those of strings $1, \ldots, r-1$.



The whole system is at rest. Consider everything below the top of string r: string r, strings $1, \ldots, r-1$ and the r particles on them. The only external forces on it are its weight rmg downwards and the tension T_r in string r upwards (the strings are light). So

$$T_r = rmg.$$

By Hooke's law, the extension of string r is

$$x_r = \frac{T_r\,\ell}{\lambda} = \frac{rmg\ell}{\lambda}.$$

So the length of the long string is

$$\sum_{r=1}^{n}(\ell + x_r) = n\ell + \frac{mg\ell}{\lambda}\sum_{r=1}^{n} r = n\ell + \frac{n(n+1)mg\ell}{2\lambda}.$$

The elastic energy in a string of natural length ℓ stretched by x is $\lambda x^2/(2\ell)$. For string r this is

$$\frac{\lambda}{2\ell}\Big(\frac{rmg\ell}{\lambda}\Big)^2 = \frac{r^2m^2g^2\ell}{2\lambda},$$

and using $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$, the total elastic energy is

$$\frac{m^2g^2\ell}{2\lambda}\cdot\frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)(2n+1)\,m^2g^2\ell}{12\lambda}.$$

The heavy rope

Ask yourself How does the first part help with a rope whose mass is spread continuously?

Cut the rope into n equal pieces. Each has natural length $\ell = L_0/n$ and mass $m = M/n$, and each piece has the same modulus λ as the rope (the modulus doesn't depend on the length of the piece). If we lump each piece's mass at its lower end, we get exactly the long string above. As $n \to \infty$ the pieces get shorter and the lumped model approaches the real rope.

Substituting into the length,

$$n\ell + \frac{n(n+1)mg\ell}{2\lambda} = L_0 + \frac{n(n+1)}{2\lambda}\cdot\frac{Mg}{n}\cdot\frac{L_0}{n} = L_0 + \frac{MgL_0}{2\lambda}\Big(1+\frac{1}{n}\Big) \;\longrightarrow\; L_0\Big(1+\frac{Mg}{2\lambda}\Big)$$

as $n \to \infty$. So

$$L = L_0\Big(1+\frac{Mg}{2\lambda}\Big),$$

as required.

You can also do this directly. Let s be the natural length of rope below a given point, so that the point supports weight Mgs/L_0. A small element of natural length δs there has tension $T = Mgs/L_0$, so it stretches by $T\,\delta s/\lambda$. Adding up,

$$L - L_0 = \int_0^{L_0}\frac{Mgs}{\lambda L_0}\,\mathrm{d}s = \frac{MgL_0}{2\lambda}.$$

For the energy, the same substitution gives

$$\frac{n(n+1)(2n+1)}{12\lambda}\cdot\frac{M^2g^2}{n^2}\cdot\frac{L_0}{n} = \frac{M^2g^2L_0}{12\lambda}\Big(1+\frac{1}{n}\Big)\Big(2+\frac{1}{n}\Big) \;\longrightarrow\; \frac{M^2g^2L_0}{6\lambda}.$$

(Directly: the element stores $T^2\,\delta s/(2\lambda)$, and $\int_0^{L_0} \frac{M^2g^2s^2}{2\lambda L_0^2}\,\mathrm{d}s = \frac{M^2g^2L_0}{6\lambda}$.)

To write this in terms of L, L_0 and λ, eliminate Mg using the length:

$$L - L_0 = \frac{MgL_0}{2\lambda} \quad\Longrightarrow\quad Mg = \frac{2\lambda(L-L_0)}{L_0}.$$

Hence the elastic energy stored in the rope is

$$\frac{L_0}{6\lambda}\cdot\frac{4\lambda^2(L-L_0)^2}{L_0^2} = \frac{2\lambda(L-L_0)^2}{3L_0}.$$

Checks

- With $n = 1$ the first part gives length $\ell + mg\ell/\lambda$ and energy $m^2g^2\ell/(2\lambda)$: a single string holding a single particle, as it should be.
- Compare a rope of the same total extension $e = L - L_0$ but under *uniform* tension: it would store $\lambda e^2/(2L_0)$. Our answer is $\frac{4}{3}$ times this, which is plausible because the stretching is concentrated near the top, and energy grows like the square of the local strain.
- Numerically, with $M = 3$, $L_0 = 2$, $\lambda = 50$, $g = 9.8$: the formula gives $L = 2.588$ and energy 5.7624; the discrete chain with $n = 100\,000$ gives 2.58801 and 5.76249, and $2\lambda(L-L_0)^2/(3L_0) = 5.7624$.

Common slips

- Taking the tension in string r as $(n - r + 1)mg$: the numbering puts string n at the top, so string r carries r particles.
- Giving each piece of rope modulus λ/n. The modulus is a property of the material and cross-section, not of the length; it is the stiffness λ/ℓ that changes.
- Leaving the energy in terms of M and g when it is asked for in terms of L, L_0 and λ.

Beyond STEP

Passing from n lumped masses to a continuous rope is the basic move of continuum mechanics: the tension becomes a function $T(s)$, Hooke's law becomes a relation between local stress and strain, and the energy becomes an integral of an energy density. The same chain of masses and springs, set vibrating, gives in the limit the wave equation for a string. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see these ideas developed.

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